

1. Firstname, lastname, Student ID:
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INDR 252 CLASS EXERCISE, February 28, 2024

Let $\{X_1, X_2, \dots, X_5\}$ be a random sample, where X_i 's are exponential random variables with an unknown parameter λ with the following PDF:

$$f_X(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

The sample has a mean of 0.6 and a standard deviation of 0.41:

1.3, 0.3, 0.5, 0.9, 0.2

Find the maximum likelihood estimator of λ .

$$L(\lambda; x_1, \dots, x_n) = L(\lambda) = \text{likelihood function}$$

$$\begin{aligned} L(\lambda) &= \lambda e^{-\lambda x_1} \lambda e^{-\lambda x_2} \dots \lambda e^{-\lambda x_n} \\ &= \lambda^n e^{-\lambda \sum_{i=1}^n x_i} \end{aligned}$$

$$\ln L(\lambda) = \ln(\lambda^n e^{-\lambda \sum x_i}) = n \ln \lambda - \lambda \sum x_i$$

$$\frac{d \ln L}{d \lambda} = \frac{n}{\lambda} - \sum x_i = 0$$

$$\Rightarrow \frac{n}{\lambda} = \sum x_i \Rightarrow \lambda = \frac{n}{\sum x_i} \Rightarrow \lambda = \frac{1}{\bar{x}}$$

$$\frac{d^2(\ln L)}{d\lambda^2} = -\frac{n}{\lambda^2} < 0 \quad \leftarrow \text{maximizing point}$$

$$\Rightarrow \hat{\lambda}_{MLE} = \frac{1}{\bar{X}} = \frac{1}{0.6} = \frac{5}{3}.$$